

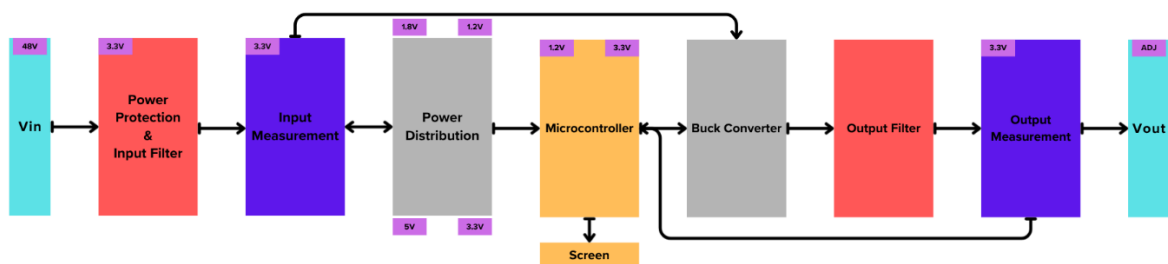
## Designing with GaN – Miniaturization and Heating of High Frequency Converter

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### First Project Design Report

**Project Purpose:** Development of educational demo board with the possibility of observing by students the heating of the system elements with changes in parameters such as frequency, load, voltage and gain.

#### Stages 1: Circuit Parts & Block Diagram



- **$V_{in}$** 
  - This part is just the power input of the circuit.
- **Power Protection & Input Filter**
  - This component protects the circuit from factors such as over/under voltage, short circuits, overcurrent, and inrush current. It also filters the input from factors such as noise.
- **Input Measurement**
  - This part measures the input voltage and current.
- **Power Distribution**
  - This part distributes the input voltage to our operating voltage levels.
- **Microcontroller**
  - This part controls and supervises all systems. It performs many tasks, such as measuring input/output voltage/current, switching the buck converter, controlling the display, and adjusting output variables.
- **Buck Converter:**
  - This part includes the buck converter topology
- **Output Filter**
  - This part creates the filter for the output after switching.
- **Output Measurement**
  - This part measures the output voltage and current.
- **$V_{out}$** 
  - This part is just the power output of the circuit.

## Stages 2: Determining Electrical Values

- $U_{IN} = 48V$
- $U_{OUT} = \text{Adjustable}$
- $F_{sw} = \text{adjustable}$
- $L =$ 
  - This value has a formula [1]:  $\frac{U_{OUT} \times (U_{IN} - U_{OUT})}{\Delta I_L \times F_{sw} \times U_{IN}}$
  - $\Delta I_L = 0.3 \times I_{OUT}$  and If our  $U_{OUT} = 12V$  and  $P_{OUT} = 100W$  then  $I_{OUT} = 8.33A$  &  $\Delta I_L = 2.5A$ 
    - Calculation factory of “0.3” is standard value.
  - Final calculation is  $L = \frac{12V \times (48V - 12V)}{2.5A \times 500 KHz \times 48V} = 7.2\mu H$
- $C_{IN} =$ 
  - We need a bulk and filter capacitors for the input capacitors.
    - Bulk Capacitor Formula:  $C = \frac{\Delta I \times \Delta t}{\Delta V}$ , If our  $\Delta I = 2A$ ,  $\Delta t = 10\mu s$ ,  $\Delta V = 1V$   
 $C = \frac{2A \times 10\mu s}{1V} = 20\mu F$
    - Input Filter Capacitors: It is aimed to create a wide-scale filter by adding **100 nF**, **1  $\mu F$**  and **3.3  $\mu F$**  capacitors to the input filter capacitors.
- $C_{OUT} =$ 
  - This value has a formula [2]:  $\frac{\Delta I_L}{8 \times F_{sw} \times \Delta V_{OUT}} = \frac{2.5A}{8 \times 500 KHz \times 0.1} = 6.25\mu F$

**NOTE:** When selecting capacitors, MLCC X7R low tolerance capacitor selections are priority.

## Stages 3: Component Selections

- MCU:
  - Texas Instruments - F28P650DK6PZPR
    - If we choose this MCU, It supports the high frequency PWM resolution. We can change the frequency and duty cycle with small movements.
    - Also this MCU has a two core. We can easily control the screen with the other core.
    - This MCU has a lots off PWM and ADC pins.
    - Price is 14.71€ per unite / 10.10€ per unite (100 Quantity)
- GaNFET:
  - The EPC2306, EPC2307, EPC2308, and EPC2367 were found to meet the desired range and specifications. Based on differences such as RDSon and size, it was decided to use **EPC2367**.
    - $V_{DS} = 100V$
    - $R_{DS(on)} = 1.2\text{ mOhm}$
    - $I_D = 101A$ , Pulsed 420A
    - $3.3 \times 3.3$
- Temperature Sensor:
  - Texas Instruments - TMP119AIYBGR
    - It has high sensitivity and small size.
- Current Measurement:
  - Current Sense Amplifier: TI INA240
- Power Protection:
  - TPS2492PWR

## References

[1], [2] B. Hauke, “Basic Calculation of a Buck Converter’s Power Stage,” Dec. 2011. [Online]. Available: [www.ti.com](http://www.ti.com)